Evaluating "Using Machine Learning to predict realized variance" by Peter Carr, Liuren Wu and Zhibai Zhang, in the Context of 2020 Market Volatility

Machine Learning II Final Project

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Introduction

CBOE Volatility Index (VIX Index)

Market measure of expected 30-day future volatility of S&P 500 Index (SPX)

- Variance swap-based pricing formula.
- Weighted combination of Out-of-the-Money (OTM) index risk-neutral options prices
- Not directly tradable
- Expensive to hedge

Carr et al. [1] propose ML calibrated alternatives to VIX, option prices as features

- [Regression I] Fit realized volatility of SPX
- [Regression II] Fit Variance Risk Premium (VRP)

Random Forests, Ridge Regression and Feed Forward Neural Network using ReLU activation function

Features

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- $(VIX/100)^2$ is a popular estimator the future 30-day realized variance of SPX
- By CBOE's definition of VIX, when there are 30-day options trading, we have

$$(VIX/100)^2 = \sum_h 2\Delta K_h e^{R_{30}T_{30}} \frac{O(K_h, T_{30})}{K_h^2}$$

- Use normalized $O(K_h, T_i)/K_h^2$ as features X
- Variance Risk Premium

$$VRP = VIX_t^{*2} - Var_t$$

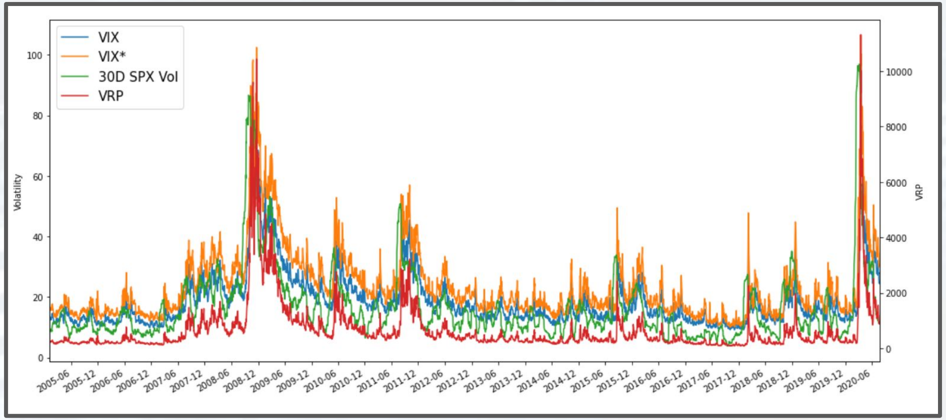
- How many strikes to consider?
    - VIX: liquid and illiquid ones
    - VIX*: $2n+1$

Data and Feature Engineering

- Bloomberg sourced Implied Volatility (IV)
  - 10, 25, 40 delta and ATM (7 strikes)
  - 30D maturity
- 1M Libor, SPX spot (and historical vol), VIX spot
- From Jan 2005 to Feb 2021
- Calculate implied strikes for Black-Scholes, e.g., for 10 delta, solve *K* in

```
@BQL("SPX Index","dropna(implied_volatility(expiry=30D,delta=25,
Call_Put = Call ))","dates=range(-20y,0d)","cols=2;rows=4070")
```

$$N\left(\frac{\log\left(S_0/K\right)+\left(R_{30}+\sigma_{30}^2/2\right)T_{30}}{\sigma_{30}\sqrt{T_{30}}}\right)=10\%.$$



Machine Learning Algorithms (Tradability Condition)

- ML Algorithms were selected to suit the condition of tradability (as in the paper).
- Index Trader needs to hedge VIX with weighted portfolio of European SPX options.
- The Constraint is that the fitting function should be piecewise linear.
- The Linear, Ridge Regression and FNN ReLU were trained on the feature data set (option prices).
- FeedForward neural network (FNN) was built with the following functional form:

$$y(x,w) = O\left(\sum_j w_j^{(2)} \cdot h\left(\sum_i w_{ji}^{(1)} x_i + w_{j0}^{(1)}\right) + w_0^{(2)}\right)$$

Here O and h are activation functions: ReLU(x) = max{0,x} on hidden layers

- Tradability Condition (Mathematical constraint) for FNN ReLU:

$$y(x,w) = \sum_{i,j} 1_{(2),j} w_j^{(2)} w_{ji}^{(1)} x_i + const.$$

- Random Forest decision tree algorithm was also trained for comparison.
- All algorithms are implemented in Python using the Scikit-learn package and PyTorch.

Machine Learning Algorithms (Comparison)

- In general, the volatility distribution has skew, is fat tailed and has the tendency to form clusters.
- Strong correlations between options features.

Ordinary Least Squares and Ridge Regression

- OLS is immune to overfitting but can have high bias.
- OLS may overestimate in low volatility period and underestimate during high volatility period.
- Ridge Regression was ran with cross validation for regularization strengths to set at lambda 0.01.

Random Forest

- Random forests handles strongly correlated features but violates the tradability condition.
- Despite being robust, it is also prone to overfitting as there are many tuning parameters.
- Configurations: number of estimators: 100 in {10, 50, 100} and max depth ∞ in ['∞, 3, 5, 10]

Feed Forward Neural Network with ReLU units

- The hidden layer has 7 nodes, same as the number of features. Repetition: 150

Results

Two Regressions

- [Regression I] Fit realized volatility of SPX (Var) $Var_t^T = f\left(\{O_t(K_i, T_j)\}\right) + \epsilon_t$
- [Regression II] Fit Variance Risk Premium (VRP) $VRP = VIX_t^{*2} - Var_t$

Features Set: 7 **O**ption prices (**T**: 30-day horizon, Deltas, **K** : 10, 25, 40)

Train and Test Dataset (~750 days training and test)

	Jan '04 - Jun '07	Jul '07 - Oct '10	Nov '10 - Apr '14	May '14 - Oct '17	Nov '17 - Mar '21
Iteration=1	Train	Test			
Iteration=2	Train	Train	Test		
Iteration=3	Train	Train	Train	Test	
Iteration=4	Train	Train	Train	Train	Test

Out-of-sample (OOS) R^2 = Ratio of predicted variance to the actual variance

- Regression II produces greater performance than Regression I
- Linear regression model outperforms non-linear models mainly because of low number of options

R^2	Regression I	Regression II
LinearRegression	-0.146	0.622
Ridge	-0.133	0.623
RandomForest_N100	-3.153	0.301
FFN_ReLU150	-	0.594

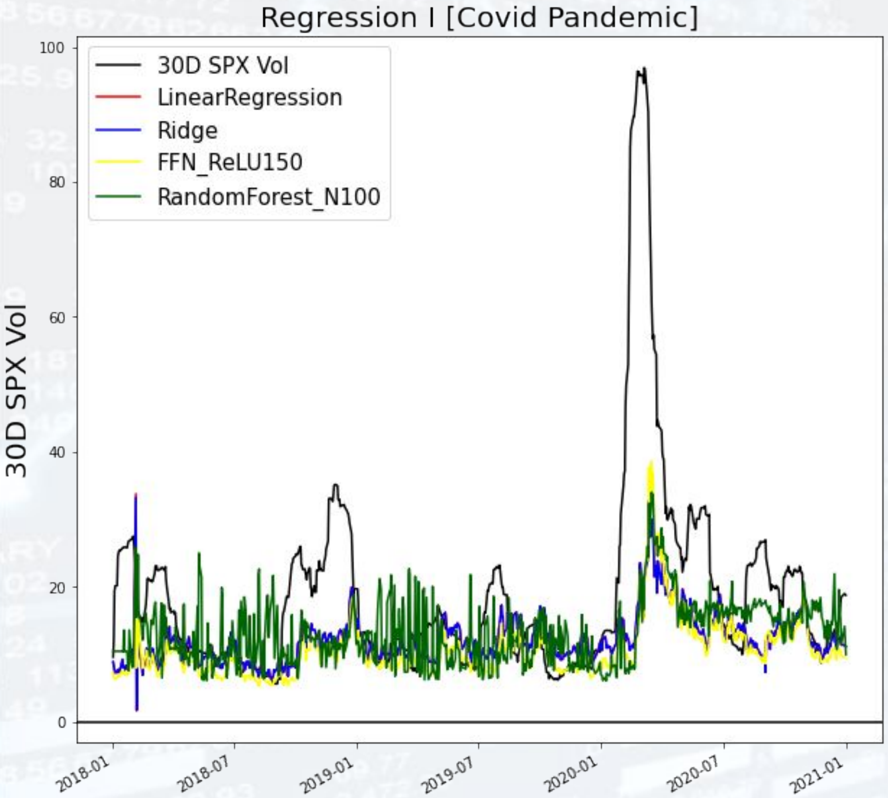
OOS R^2

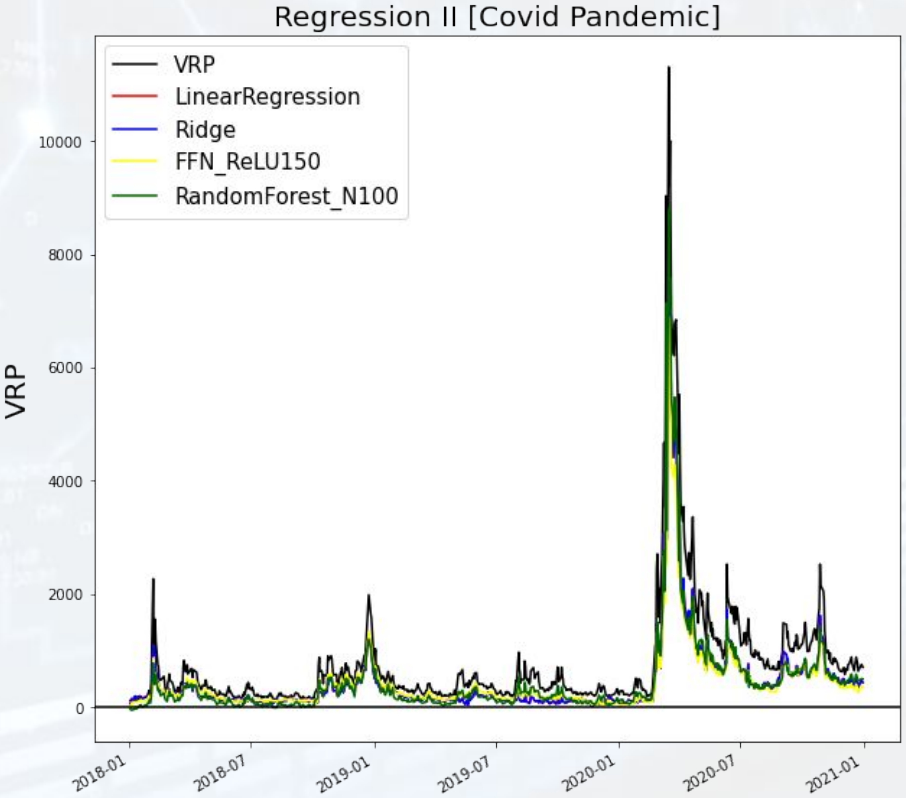
$$R^2 = 1 - \frac{\sum_i (y_i - p_i)^2}{\sum_i (y_i - \overline{y})^2}$$

Source: Shutterstock

Results - Plots

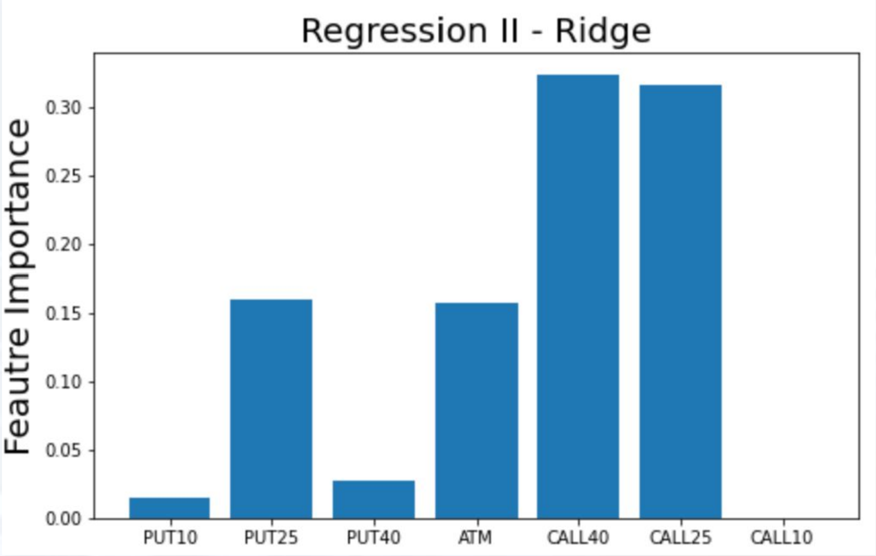
Below two graphs show model predictions for both regressions during 2020 Coronavirus Pandemic





Results - Feature Importance

- Feature Importance is measure of percentage improvement to the OOS (volatility) predictions
- Mean-Decrease-Accuracy (MDA) (scikit's Permutation Importance) was used to compute feature importance
- Two patterns emerged:
  - For both Puts and Calls, the more they were out-of-the-money, higher their importance features were
  - Calls were more important than Puts in Regression II



Conclusion

It is possible to create an index which has predictive power similar to the VIX using ML methods on liquid options

- Despite reduce option count, similar results to Carr et al.
- VRP is much more “fittable”
- Ridge Regression and FNN with ReLU fit the historical data well
- FNN with ReLU meets tradability requirements
- Calls are more important than puts (at least in Regression II)
- Performance during Coronavirus period is good

Future Extensions

- More option strikes likely give better fits
- More hyperparameter exploration, RNNs?
- 60D, 90D option expiry
- Other features (historical volatility, overnight rates etc.)

Thank You For Listening

Source: Shutterstock

Thanks to:

Professor Reinhart

Class TAs

Classmates

Teammates

References

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